Frontiers of Architectural Research (2021) **10**, 117–133

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Research Article

Architectural survey of historical buildings: The orders of classical architecture in the Baptistery of Florence

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Received 25 March 2020; received in revised form 28 June 2020; accepted 3 July 2020

KEYWORDS
Architectural survey;
Recovery *rilievo*;
Photogrammetric survey;
Architectural conservation;
Classical architecture orders;
Architectural symbolism

Abstract The survey of historic buildings is very important for architectural conservation. An integral survey allows perpetuating the building historical memory, deciphering it for future generations, and recovering it in case of accidental loss. The Baptistery of Florence survey was done with four measuring methods: the "direct" one, which employs traditional measuring tools; the "indirect" one, which expedites the dimension-gathering process with more precision; and the "photogrammetric" one, which uses snapshots to facilitate the representation process with computers. An innovative measuring concept, known here as the "recovery survey", synthetizes graphical reality so that irregularities—due to the project materialization and deterioration over time—disappear; therefore, retrieving the building original design. Columns with their bases, shafts and capitals, as well as entablatures with their architraves, friezes and cornices were under study. The three Greek orders (Doric, Ionic and Corinthian) together with the two roman orders (Tuscan and Composite) unveiled the classical architecture significance. These constituents, which reoccur inside and outside the baptistery, were measured as part of these holistic survey and recovery process, to achieve the objective of this research study: the recording of this world historical building through an integral survey and *rilievo* to decode its significance and symbolism.
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Peer review under responsibility of Southeast University.

https://doi.org/10.1016/j.foar.2020.07.002
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1. Introduction

The conservation of architectural monuments classified as tangible cultural heritage opens a new perspective in the field of historical research in architecture. The preservation of a constructed heritage of any city anywhere in the world is important. To do this, we first have to make an inventory of what exists and what we need to perpetuate in documents. We must carry out not only an architectural survey but also a holistic survey, called *rilievo*, taking the word as a borrowing from the Italian language, of the monuments categorized as national patrimony. The difference between survey and *rilievo* is quite large. The architectural survey is a process through which the architectural components are measured for acknowledging the shape and dimensions, while the second one goes far beyond that. The first one is a process of descriptive geometry that aims to represent a manufactured object in the majority of cases and architectonic. The second one refers to several processes that respond to the overall purpose of knowing everything about the building under study, not only its shape and dimensions. When we speak of architectural *rilievo*, we also mean the study of the building past and evolution, its architectural interventions, and its characteristics not only formal but also functional and historical. Architectural *rilievo* also takes into account the monument color, material, disposition, and orientation, among other physical and nonphysical features, such as interpretation and perception as the result of a *gestalt* analysis, and others (e.g., its symbolic load, meaning, and origin of a given form). This article aims to understand the origin and reason of shapes through the architectural survey and *rilievo* and discover and analyze the symbolism load behind all the forms found in the monument of study.

The monument that has given me the opportunity to put into practice my knowledge about architectural *rilievo* and survey is precisely the "Baptistery of Saint John in Florence"—the "Battistero di Firenze" or the "Bel San Giovanni". This personal interest of knowing more about this monument has been the fundamental stimulus that ultimately allowed me to investigate and publish a series of articles with a common subject, proposed from the start of my career itself. Furthermore, this experience has provided me with skills to translate the same techniques of architectural investigation to other existing monuments in our country and in particular in a project that is now being carried: the *rilievo* of the Main Church of Zaruma, which is a beautiful and picturesque village in the highlands of El Oro province in the south of Ecuador.

2. Materials and methods

The architectural *rilievo* of the Baptistery of Florence was carried out under my supervision with the collaboration of five other people: two working groups of three people each one, that is, two persons for handling the basic measuring tools and a third person for recording these measures on paper. We began with a set of existing blueprints to not only mark and delineate areas of work but also divide the various areas and include them within a proposed work schedule. We used the following mechanisms out of the several known architectural survey and *rilievo* methods:

The first method of architectural survey called "direct" runs with the help of traditional tools of measurement, such as flexometers, measuring tapes (dual-scale, roll-up, self-retracting plastic, or metal measuring tapes), calibers, mechanical vernier scales, and micrometers. Many technological and measuring devices behind the direct method taught us how to use these casual measurement tools to achieve the optimal accuracy as much as possible.

After bidimensionally measuring everything (planimetric survey), we checked the dimensions by triangulation. The partial and total heights (altimetric survey) and the window and wall thickness (volumetric survey) were taken with the help of less traditional and more technological instruments, such as theodolites, micrometers, and digital altimeters. This second method, known as "indirect", facilitates and accelerates the collection of measures. The degree of accuracy is great when we use these technological tools instead of the conventional ones. However, this method involves the use of costly rental equipment as electronic and digital theodolites.

We also used a third method to lower the budgetary costs. This method is innovative but not new because it emerged in the 19th century almost simultaneously with the advent of photography known as "photogrammetric", which uses high-resolution frames captured with professional cameras with high-quality lenses taken from certain points of views. We straighten these images and convert them into facades with parallel vanishing lines, such as elevations, completely perpendicular to the point of view with the help of a computer and specialized software. Some length measures were taken, which were long enough to be considered representative, and used them as a reference on the frames. Then, we scaled the frames according to these measures. Finally, we obtained the rest of the lengths with the same scale of all the remaining elements of the architectural monument, especially the facades. In this way, we manage to speed up the survey of floors, interior, and exterior facades. The representative elements of architectural classical orders, such as columns and entablatures, are found. This method allowed us to survey areas with difficult access, such as the *lantern* that crowns the baptistery octagonal dome.

Photogrammetry was an indispensable tool for the elaboration of detailed architectural surveys, such as those of capitals and architraves, which would be almost impossible to register with either the direct or the indirect methods, unless the drafter has adequate access and is an artist of representation. Photogrammetry is useful not only for architectural conservation but also for a wide range of other aspects, such engineering surveys.

This third method was less expensive than the second one even though we required state-of-the-art photographical equipment, powerful computers, and sophisticated CAD programs for an accurate architectural representation.

A fourth method of survey is called "3D scanning", which still remains experimental, not because of the lack of technological tools of last generation but due to the incapacity of these instruments to meet the architects' needs to survey images from several points of view. Various types

of 3D scanners are available, from those residential and small portable to those professional and large-format ones; however, all of them have the same disadvantage: the resulting image of this scanning is holistic and cannot be decomposed in parts without the help of an appropriate software; in addition, they cannot register architectural components that cannot be seen, such as those behind points of difficult access. The 3D scanners are useful to document pieces of art of a certain size, such as sculptures. This method is still insufficient for architectural monuments. However, we can scan the entire facades of buildings or full-in-length street facades from a fixed point. We cannot scan what is behind these facades. The resulting images are not decomposable in their parts, for example, the complex modeling of a window in three dimensions with its jambs, girdles, struts, edges, locks, glass, and sills. All the work to break down an architectural object into its constituents does not justify the use of large-format scanners for any architecture survey. This method was not used in the process of this particular survey project because of the constraints that it presents.

From an organizational point of view, the resulting information of this project was divided into three phases: a first textual part, which explains the general characteristics of the object under study (classical orders of the baptistery) and ends with several original contributions about the constituent elements of the Baptistery of Florence (bases, columns, capitals, architraves, friezes, and cornices); a second photographical part, which supports the entire textual material on the basis of monochromatic images; and a third graphical part, which enhances the previous part through representations that are the result of partial and integral architectural surveys.

The first textual part was further divided into other four parts to structure them in a hierarchical way: the *lemma*, *text*, *captions*, and *bibliographic citations*. The *lemma* is the key word that starts each article referring to each part of the baptistery architecture (e.g., base) and identifies each constituent element of the classical orders. The *text* is the heart of the article and contains approximately an 80% of the total original information of each article, that is, the knowledge resulting from the symbolic load analysis identified through the geometric and organic elements found within the three classical orders. The second photographical part highlights the formal characteristics of the architectural component through a photographic material. This part is directly related to the core of the text described in the preceding part. The size of the photographs demonstrates the importance of the architectural details, depending on the symbolic-formal aspect. The photographs are numbered with their corresponding captions. The photographs were taken by professional photographers with large-format cameras to achieve high-resolution photograms. We used slide and paper photographs printed in color and scale of grays. However, the slides and prints were scanned to digitally treat and then make them useable for the photogrammetric survey with the adequate computer software. The important role that the photographic shots played in this investigation, representing a vital part in the understanding of the results, is indisputable.

The third graphical part amplifies, clarifies, and completes the symbolic and formal characteristics of the element treated in the article. In this part, the survey and *rilievo* have a predominant role of understanding *relievo* not only as the mechanical registration of the elemental measures but also, and most important of all, the recording of the background information that comes along with the architectural survey, such as the data about the materials used, the historical evolution of the elements, the color of the finishing, the individual perpetrators of these constructions, and the current state of deterioration. We used three types of CAD software for the elaboration of the graphic representations. Despite the availability of computer means for this specific purpose, the automated drawing was a great challenge due to not only the geometric and organic complexity of the architectural details but also the degree of precision that was set as an initial objective. As a point of departure, many representations were carried out from the photograms themselves to later on proceed with a *rilievo* using the CAD rasterized images. The photogrammetric *rilievo* would not have been possible without computer-assisted drawing. The graphic images have been numbered and described by their own corresponding caption notes. The criterion for the graphical caption—note writing is the same as that for the photograph caption—note writing.

We had introduced a new type of conceptual graphical representation (survey for recovery) to achieve this project. In this type of survey, the images retrieved through the various processes of measure taking are initially based on their shape and graphic reality. Afterward, the images are systematized and synthetized to ensure that the imperfections—among other incongruences that are product of the project materialization and difficult workmanship—disappear, such as the repeating elements of the inlaid marbles of the pavement or the ova, darts, and arrows of the cornices and entablatures. The survey for recovery considers the differentials and inconsistencies of the baptistery architectural components, which by the way are understandable due to the manufacturing and similarities; therefore, they are represented exactly alike, in absolute terms. The capitals of the columns, which are theoretically and conceptually symmetrical, but in reality are not, due to the reasons already mentioned above, are considered symmetrical, that is, just as by the time when they were originally conceived on the drawing board.

At this point, we can speak of a survey for recovery, considering that the repetitive rhythm of identical items or of symmetrical components is transformed into a sequence of equal and symmetrical elements in absolute terms. This type of survey does not alter the reality but rather seeks to approach the authenticity of the original conceptual project. The survey for recovery not only represents the vestige of the primitive ideological project but also favors the survey that seeks to represent what existed without the incongruences due to manufacturing and the state of current degradation due to the passage of time and use of the architectural monument. These two types of survey are not antagonistic, but rather complementary. The two methods can perfectly coexist. For example, we tried to record the project changes from its primitive interpretation, passing

through all the historical interventions until its current state, the Baptistery of Florence survey for recovery. This situation resulted in a virtual restoration that allowed us to reveal the greatness of the architectural monument at the very moment of its conceptualization, just as if we had done a trip back in time, thereby removing the relentless effects of degradation because of the monument centuries of existence.

3. Results and discussion

3.1. Architectural classical orders

The orders of classical architecture have been reproduced since their conception itself on the basis of the classical orders of Greek and Roman architecture recaptured by the architecture of the Renaissance and subsequently reinterpreted in the modern movements of architecture, such as postmodernism. Classical-architectural orders have a visible influence on the evolution of art through their different historical architectural manifestations and styles. The Baptistery of Florence is not an exception. The three Greek orders' par excellence (Doric, Ionic and Corinthian), in addition to the two roman ones, derived from the previous three (Tuscan and composite), is the manifestation to a greater or lesser degree of a quasi-perfect symbiosis, which feeds the symbolic value of the Baptistery of Florence (Verdon, Timothy. 2017. *Il grande museo del Duomo di Firenze*). These five orders must be decomposed into their constituent elements to effectively interpret their symbology: columns with their bases, shafts, and capitals; entablatures with their arquitrabes, friezes, and cornices; and all the other subcomponents of these six major elements.

3.2. Columns: shafts and types of columns

The word "column" comes from the Latin "columna", from *excellere* (be tall). The column is the vertical architectural element composed of circular sections with a primary function of supporting other load-bearing structural elements (e.g., walls, arches, and naves) and a secondary function of decorating edifices or commemorating a special event. The column consists of three main parts: base, shaft, and capital, with different proportions, shapes, and decorations depending on the antiquity, style, and cultural field: Doric, Ionic, Corinthian, Tuscan, and composite; made of stone, marble, and granite; and honorary and votive.

In a figurative language, the column symbolizes support and stands for moral back up, as in the expression: the eldest son is the family column. In a generic sense, "the column not only signifies uphold but also personifies force and vigor by itself and eventually embodies severe and austere loneliness." (Cairo, Giovanni. 2001. *Dizionario Ragionato dei Simboli*).

"Columns are the essential elements of architecture, structurally speaking; columns are the main load-bearing vertical elements: they represent the backbone of construction and unite the different levels, ensuring the building strength. Moving them means threatening the entire building; for this reason, they are often considered as the whole." (Chevalier, Jean e Gheerbrant, alain. 1996. *Dictionnaire des symbols*). A biblical episode can be counted as an example to everything that has been said in this paragraph: Samson, prisoner of the Philistines, broke down the columns of the temple, scared away their enemies, died with them, and gave victory to his people. "Then Samson reached toward the two central pillars on which the temple stood. Bracing himself against them, his right hand on the one and his left hand on the other, Samson said, 'Let me die with the Philistines!' Then he pushed with all his might, and down came the temple on the rulers and all the people in it. Thus he killed many more when he died than while he lived." (Judges 16: 29—30).

In architecture, the columns are not only the structural element with support function but also have a rich symbolic meaning. The column with its base and capital symbolizes a tree: the base represents the roots, the shaft is the trunk, and the capital is the foliage. The column brings the building to life and to everything that it stands for. The column precisely takes its shapes from the tree. For example, most Egyptian columns portray the typical plants of the place as the palm tree and papyrus plant.

In the Hebrew and Christian traditions, the column assumes a cosmic and spiritual symbolism; it supports the high and serves as a connection between the latter and the low. The column can be consistently related to the cosmic tree and the tree of life (Chevalier, Jean e Gheerbrant, alain. 1996. *Dictionnaire des symbols*). Elizabeth Rees considers that the column, for Christians, represents the cosmic trunk of the world, which is the cross (Rees, Elizabeth. 1994. *Simboli Cristiani e Antiche Radici*).

The columns are often located on both sides of a temple entrance, as in the case of the Baptistery of Florence; the Pisans donated two porphyry columns to the Florentines in early in the XII century as appreciation for the strong contingent of Florentine troops to the victorious Pisan expedition against the Balearics in year 1113 (Busignani, A. e Bencini, R. 1988. Le Chiese di Firenze. Il Battistero di San Giovanni). (Photo 1).

According to the Bible (*Job: 9, 6*), on the Day of Judgment, only God has the power to break down the columns that hold the world. From this point of view, we can conclude that columns are a symbol of universal support that God alone is able to demolish. The columns within the baptistery have the symbolic role to sustain the dome, which in turn represents the universe created by God. The mosaics of the inner dome are distributed in seven registries that represent 1) plant-shaped illustrations and reproductions of fantastical back to back animals in pairs, 2) angelic hierarchies and Creator, 3) stories of Genesis, 4) stories of Joseph the Hebrew, 5) stories of Jesus Christ, 6) stories of Saint John the Baptist, and 7) final judgment.

All columns, beginning from the lowest (first internal registry) until the small columns of the double-lancet windows (second internal registry or *emporas* [women's galleries]) to then continue to those represented in the dome mosaics—as separation of the different biblical episodes—converge toward a point past the top of the dome drum in the direction of Heaven as a single axis from the terrestrial world to Paradise (Photo 2).

The symbolic link between column and tree is obvious. The column base, with its rounded shape and diameter,

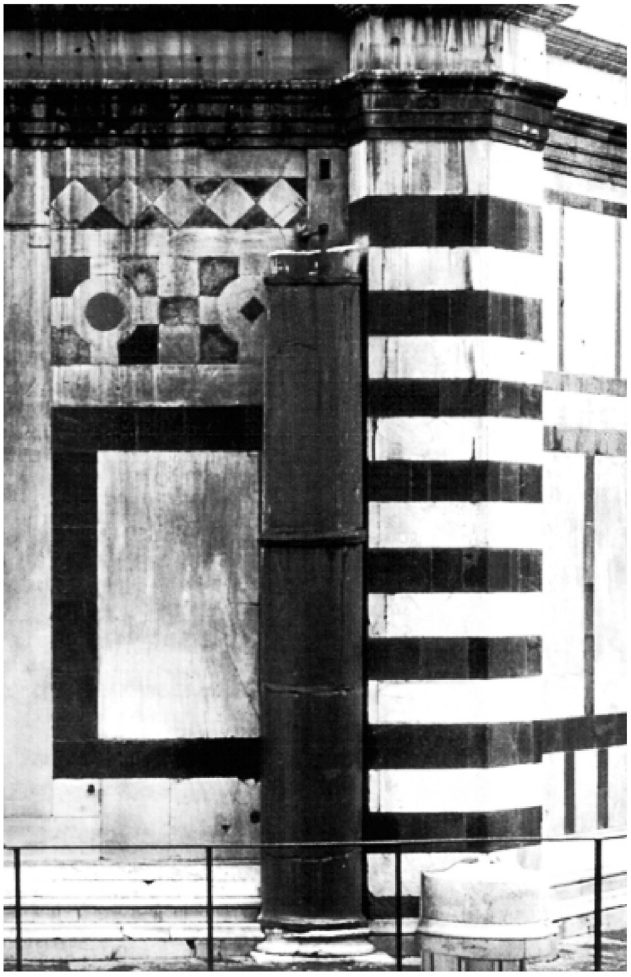


Photo 1 Porphyry columns that frame the entrance (Door of Paradise) of the Baptistery of Florence donated by the Pisans for the Florentines in the 12th century.

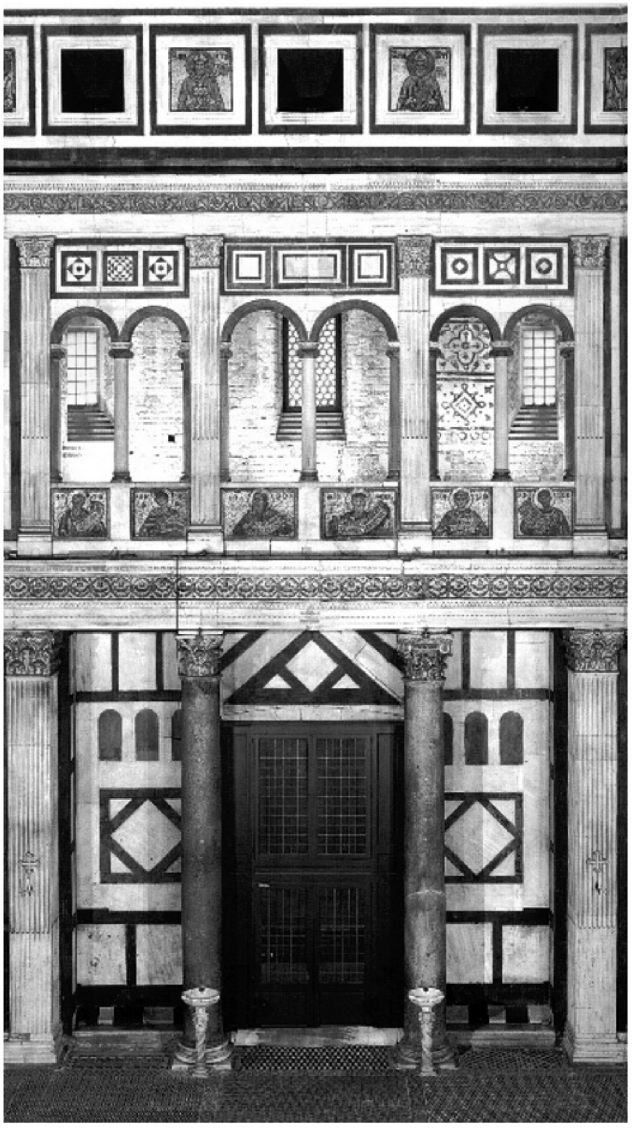


Photo 2 Columns and small columns of the first and second internal orders that represent the support of the universe created by God and the axis of the world.

greater than the shaft, indicates the tree baseline together with the roots. The column shaft (sometimes with entasis [with a diameter widening in the lower zone of the shaft for aesthetic and visual reasons] and fluted [with regular concave grooves along its height]) is narrower than the base and much more lengthened and represents the trunk of the tree. The capital, which is often loaded with vegetable decoration (such as the acanthus leaves), symbolizes the foliage of the tree. This analogy between the column and the tree has always existed in the different cultures. For example, the ancient Egyptians built columns imitating the forms of their native plants, such as the palm tree or papyrus plant. The column of San Zanobi, in front of the north gate of the baptistery, represents a tree and recalls the homonymous miracle that took place on 26 January 1429 (Sinisgalli, Rocco. 2003. *Il Battistero di Firenze*). "While the Christian community would transfer the mortal remains of San Zanobi from the cathedral of 'San Lorenzo' to that of 'Santa Reparata'—the old church that existed in the place of the current main cathedral of 'Santa Maria del Fiore'—, in front of the baptistery north face, a dry tree, which was slightly touched by the coffin, greened" (Verdon, Timothy curated by. 1992. *Alla Riscoperta di Piazza del Duomo in Firenze*) (Photo 3). The column-tree analogy is evident in the mosaicked column representations that divide the various figurative episodes inside the baptistery dome (Photo 4).

The internal and external columns of the baptistery are essentially of a single type with some variations. The columns of the first internal registry are with entasis and plain, except the one with entasis but fluted (from the end of the column lower third up to the top) and tongued (with concave grooves filled up with convex moldings from the column base up to the end of the lower third), which frames the right side of the door known as "Gate of Paradise" with an attic base (e.g., the rest of the columns) and a Corinthian capital (e.g., the rest of the columns, with the exception of the two capitals of the columns that frame the east side gate or "Gate of Paradise" and the capital of the column that frames the right side of the north gate; these three capitals are composite in type) (Figs. 1 and 2). The



Photo 3 Arboriform column that recalls the miracle of San Zamboni.

small columns of the second internal order known as "*emporas*" or women's galleries are slightly with entasis and plain, with an attic base and an ionic capital surmounted by a latastrum (Fig. 3).

We must take note of the dimensions of the columns and their relationship with other structural elements of the Baptistery of Florence (e.g., walls), or the relationship between the elements of the columns themselves (e.g., bases), the shafts, and the capitals, or even other types of relationships similar to that between columns and intercolumnial spaces. Giovanni del Rosso was one of the first erudite of the baptistery who was interested in this argument, referring mainly to the columns of the first internal order, which in effect are those that adapt to an analytical-descriptive study that uses analogies on the basis of previous studies (Accidini, Cristina. 1997. *Il Battistero e il Duomo di Firenze*. Ed. Illustrata). "These isolated columns are detached from the wall in the width of their diameter, and form three equal spaces, enclosed by pilasters of the same order but grooved, which are coupled at the angles leaving between them a moderate space, so that the abacuses of the capitals do not combine between them. The pilaster-to-pilaster space, which



Photo 4 Arboriform columns that divide the different episodes of mosaics inside the baptistery dome.

encloses the intercolumnial spaces, is 16 arms-length and a third. This, having been divided into 14 parts, each of them an arm-length and a sixth, which was the diameter

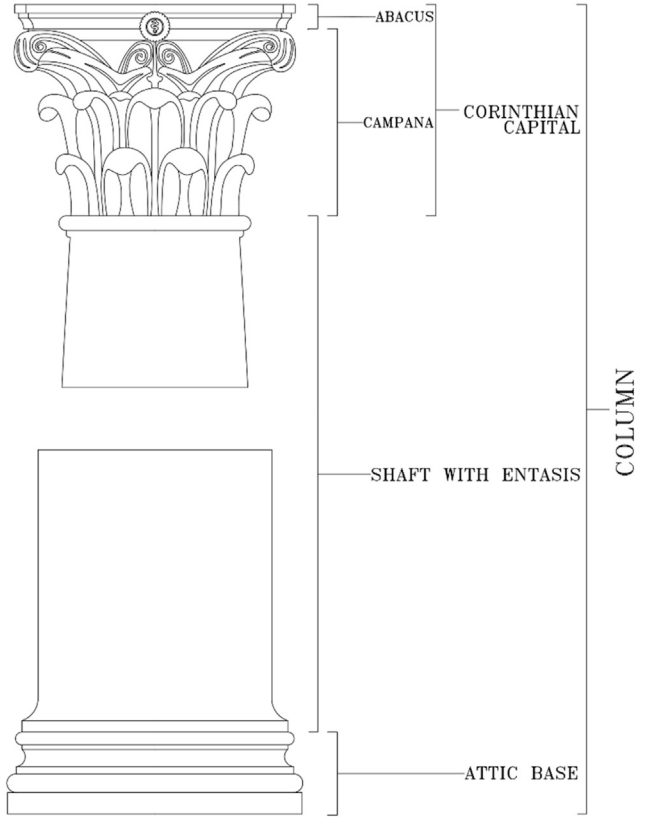


Fig. 1 Representation of a typical column of the first internal order: attic base, shaft with entasis, and Corinthian capital.

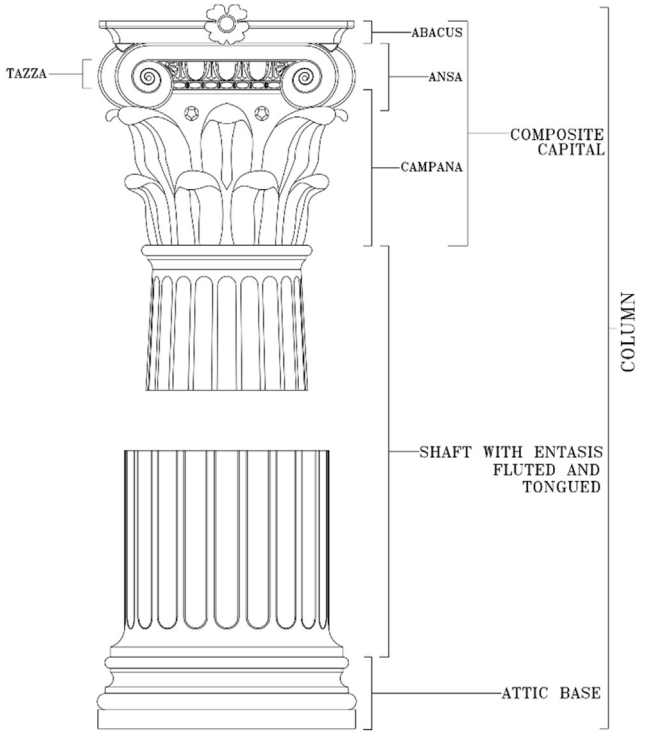


Fig. 2 Representation of a column that frames the right side of the Gate of Paradise of the first internal order: attic base, shaft with entasis, grooved, and fluted, and composite capital.

of the columns found, two of these parts are occupied by the columns themselves, and there are four for each of the three intercolumnial spaces, one after the other, without taking into account the many inaccuracies of the construction. The shaft of the columns is 9 diameters high; their capitals are one diameter high, as described by Vitruvius; and half of the capital is as high as column base, always excluding some insignificant alterations. The columns, with their relations, reach the height of 10 diameters and a half. For this reason, this height, being one of the greatest known of antiquity, suggests with certainty the idea of having served for a work carried out much later to Octavian times, as evidenced by a manifest indicative of the weak stalk of the corinthian capitals, except for three of them that are compound, but treated in the same way." (Del Rosso, Giovanni. 1820. *Ricerche Storiche architettoniche sopra il Singolarissimo Tempio di San Giovanni*).

The elements, such as external columns and wall-mounted pilasters, can be divided into two types: those of the main sides that contain the three gates (north, south, and east sides) and those of the secondary sides without gates (northeast, northwest, southeast, southwest, and west [scarcella]). The adjoined columns of the first order of the main sides are entasised and have an attic base (except for that column that frames the left side of the north gate with a Corinthian base and a double central

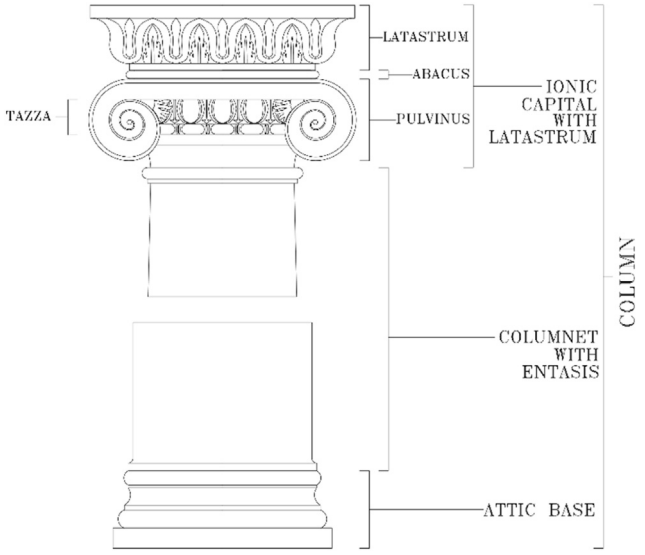


Fig. 3 Representation of a typical column of the second internal order (matronea): attic base, shaft with entasis, and ionic capital surmounted by a latastrum.

astragalus) and a composite capital (Fig. 4) and (Fig. 5). The octagonal columns of the second order, which are not entasised, also have an attic base and a composite capital (Fig. 6).

Wall-mounted pilasters without entasis of the first order of the secondary sides have an attic base and a Corinthian

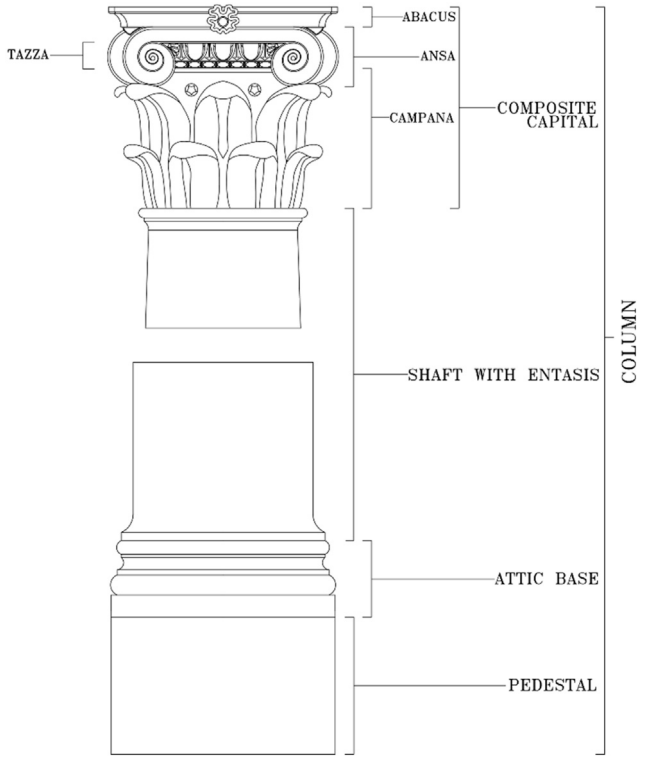


Fig. 4 Representation of a typical column of the main sides of the first external order: attic base, shaft with entasis, and composite capital.

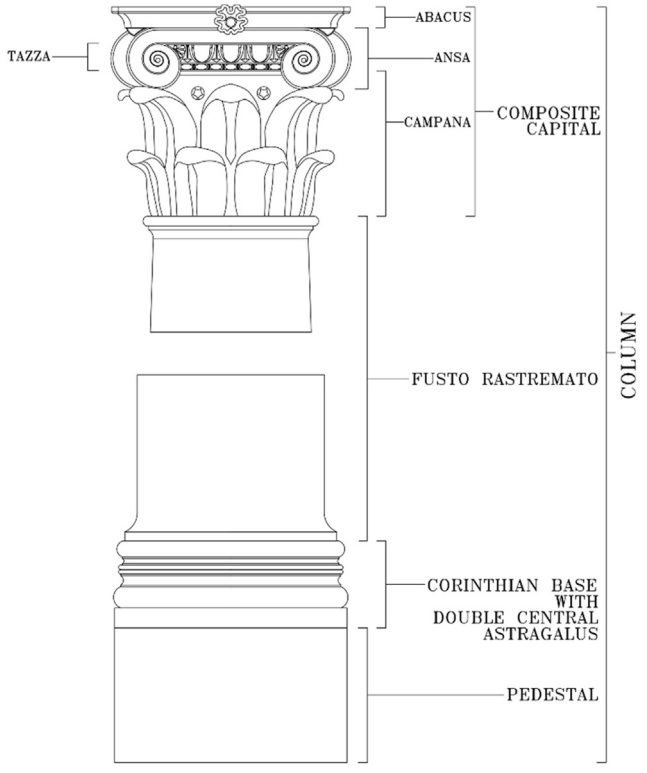


Fig. 5 Representation of the column that frames the left side of the north door of the first external order: Corinthian base, shaft with entasis, and composite capital.

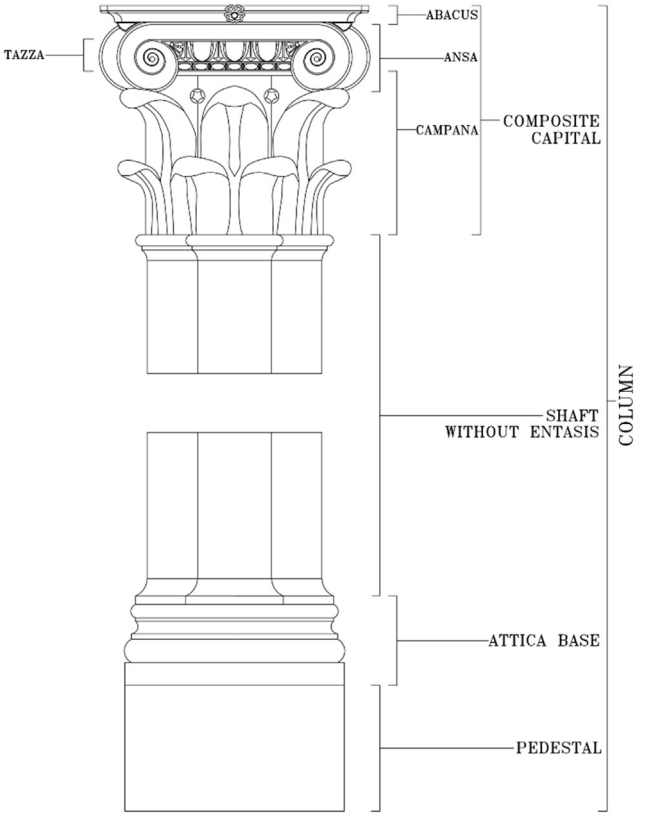


Fig. 6 Representation of a typical octagonal column of the main sides of the second external order: attic base, shaft without entasis, and composite capital.

capital (Fig. 7). The octagonal columns of the second order have the same characteristics (Fig. 8).

Columnet-pilasters and pilasters of the second external order can be classified according to the side to which the windows belong (main and secondary sides) and depending on the position of the windows inside the same sides (central and lateral sides).

With regard to the central windows of the main sides (north and south-east sides with the exception of the scarcella side), the columnet-pilasters have an attic base, a shaft without entasis (with the exception of the columnet-pilasters of the east side that have a light entasis) and diagonally grooved, and a composite capital (Fig. 9). The lateral windows have wall-mounted pilasters with an attic base, a plain shaft without entasis, and a Corinthian capital. The lateral windows on the east side have columnet-pilasters (as in the previous case: an attic base, a shaft with entasis and diagonally grooved, and a composite capital) instead of the wall-mounted pilasters.

With regard to the central windows of the secondary sides (northeast, northwest, southeast, and southwest), the columnet-pilasters have an attic base, a plain shaft with a lightly entasised length, and a composite capital, except for the central window on the southwest side, which capitals are ionic instead of composite. The lateral windows have wall-mounted pilasters similar to those of the lateral windows of the main sides; they have an attic base, a plain shaft without entasis, and a Corinthian capital (Fig. 10).

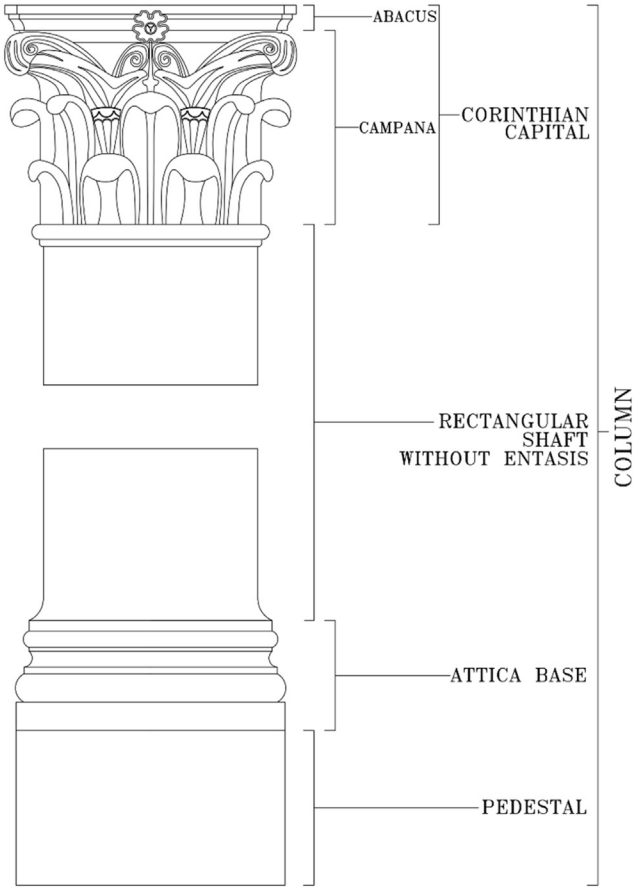


Fig. 7 Representation of a typical lesene-pilaster of the secondary sides of the first external order: attic base, shaft without entasis, and Corinthian capital.

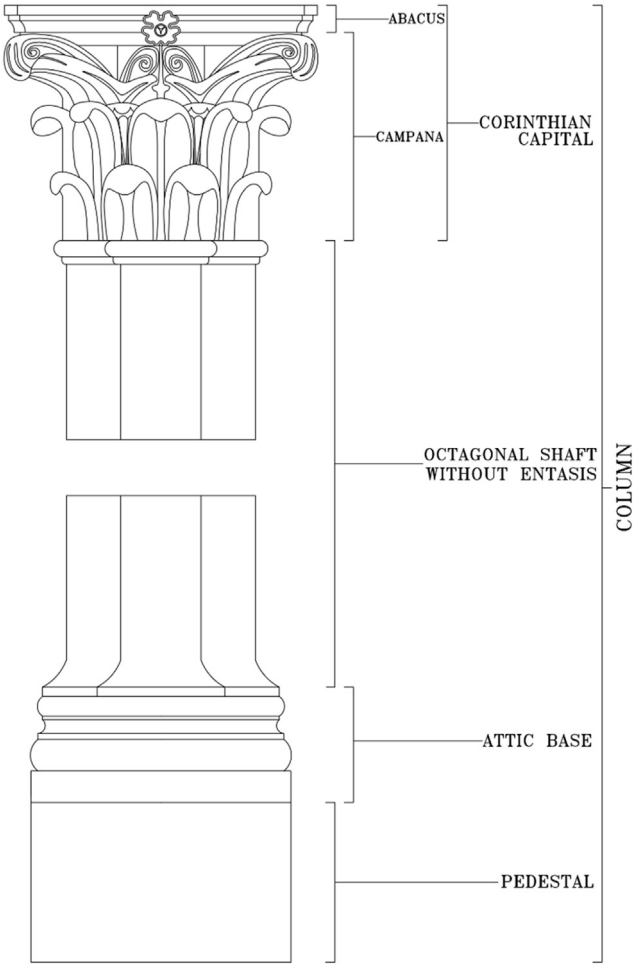


Fig. 8 Representation of a typical octagonal column of the secondary sides of the second external order: attic base, shaft without entasis, and Corinthian capital.

The two right capitals of the lateral windows on the southwest side are composite but not Corinthian.

3.3. Bases: constituent parts and types of bases

The structural function of the base symbolizes the principle or foundation of any architectural work in the general terms possible. The base is the architectural element that is directly in contact with the ground and serves as support for a building, a statue, or a column. Moreover, the base is a support or bottom of something.

The base of the column together with the shaft and the capital symbolize the tree of life: the base represents the roots, the stem is the trunk, and the capital is the foliage. This perspective is a pragmatic and particular way of seeing the column and its constituent parts. "The base of a column is considered by all peoples as the main symbol of beginning." (Ronchetti, Giuseppe. 1979. *Dizionario Illustrato dei Simboli: Simboli, Emblemi, Attributi, Allegorie, Immagini degli Dei, ecc.*).

Bases have many different types, but one of the most common is that internally or externally used in the Baptistery of Florence, that is, the attic one. This base is the attic-specific form of the ionic base, which consists of a scotia or trochilus between two toruses, sometimes without a plinth. The upper torus is in a lower position and less protruding.

The two types of bases found in the Battistero di San Giovanni in Firenze are as follows: attic (Photo 5) and Corinthian (Photo 6).

The attic bases are found inside the baptistery in the columns of the first order, the wall-mounted pilasters of the second order of the matronea, and the columnets that divide the double-lancet window of the same matronea (Fig. 11).

The attic bases with small variations are found in the columns and wall-mounted pilasters of the first order, in the octagonal columns of the second order, and in the wall-mounted pilasters of the attic. Some external bases present differences. For example, the base of the left column of the first order of the east side increases one astragalus between the scotia and the lower torus and one more astragalus between the same scotia and the upper torus (Fig. 12).

On the secondary sides of northeast and northwest, the bases of the wall-mounted pilasters of the first order present a strong simplification; its profile is reduced to that of a plinth and a torus, substantially similar to the Tuscan base (Fig. 13).

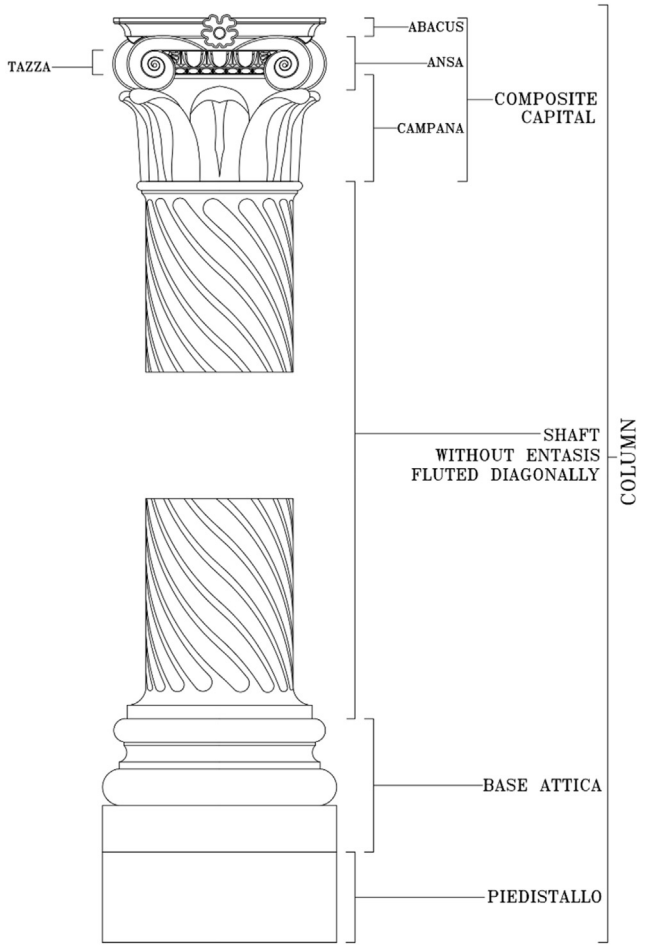


Fig. 9 Representation of one of the small column-pilasters of the central window of the south side.

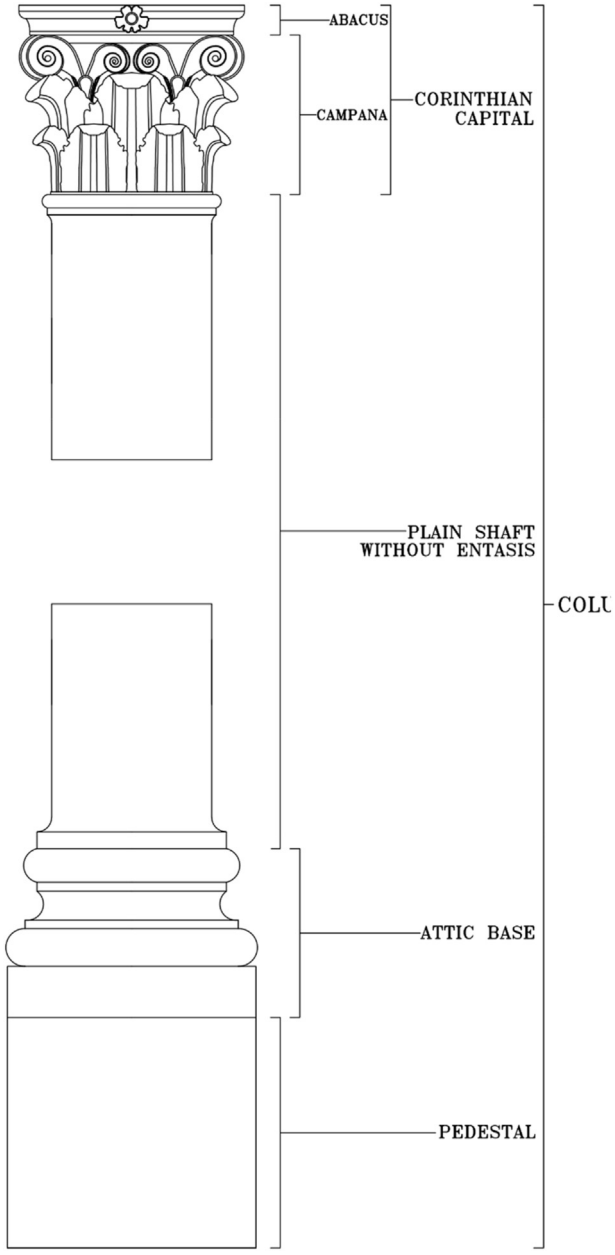


Fig. 10 Representation of one of the lesene-pilasters of the right side window of the southeast side.

The Corinthian bases are composed of a plinth, a lower torus, a lower scotia with its fillets, a double astragalus, an upper scotia with its fillets, and an upper torus. These bases are found with a small variation in the cantonal pilasters of the first order. This precious base, which is the most complex of classical antiquity, is found in the cantonal pilasters of the first order of the northeast and southeast sides (Fig. 14). The remaining pilasters have a single astraglus higher than usual instead of the double astragalus between the upper and lower scotiae (Fig. 15).

The attic bases are found in all the columns with the exception of the Corinthian column with a double central astragalus that frames the left side of the gate of the first order of the northern leaflet. The bases of the columnet-pilasters and of the wall-mounted pilasters that decorate the windows of the second external order are all attic, similar to the first variant without astragali.

3.4. Capitals: constituent parts and types of capitals

In the field of the trilithic construction system, the capital is that monolith that not only crowns an upright element (e.g., a column, a pilaster, a post, or a lesene [a pilaster strip]) but also serves as support for an architrave. The capital originates in the most elementary geometric forms, as can be seen in the evolutionary transition from the abacus or the dice to more complex and decorated forms, such as volutes, phytomorphic and zoomorphic ornaments, or mixed solutions.

The symbology of the capital is closely linked to that of the column that evokes the shape of the tree. The capital symbolizes the foliage. Many types of capitals have vegetal decoration.

A different capital meaning is that mentioned by Gioseppe Ronchetti. Just as the base of the column symbolizes the beginning, the capital symbolizes the "end because it crowns and ends the column" (Ronchetti, Giuseppe. 1979.



Photo 5 Attic base of one of the columns of the inside of the baptistery.



Photo 6 Corinthian base of two cantonal pilasters of the inside of the baptistery.

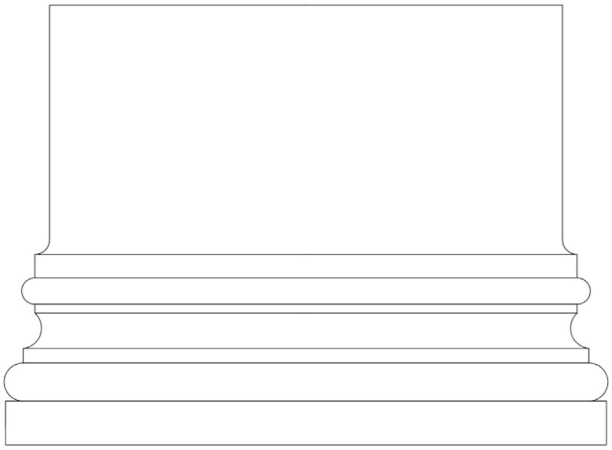


Fig. 11 Representation of the first variant of the attic base without astragalus.



Fig. 12 Representation of the second variant of the attic base with two astragali.

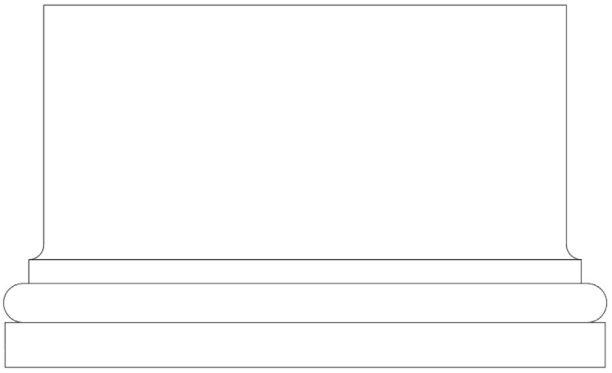


Fig. 13 Representation of the third variant of the attic base with a single torus.



Fig. 14 Representation of the first variant of the Corinthian base with a double astragalus.

Dizionario Illustrato dei Simboli: Simboli, Emblemi, Attributi, Allegorie, Immagini degli Dei, ecc.). Ronchetti also confronts these two concepts of principle and end (base and capital of the column, respectively) by symbolically relating them to the concept of "alpha and omega" (a common biblical symbol) (Ronchetti, Giuseppe. 1979.



Fig. 15 Representation of the second variant of the Corinthian base with a simple astragalus.

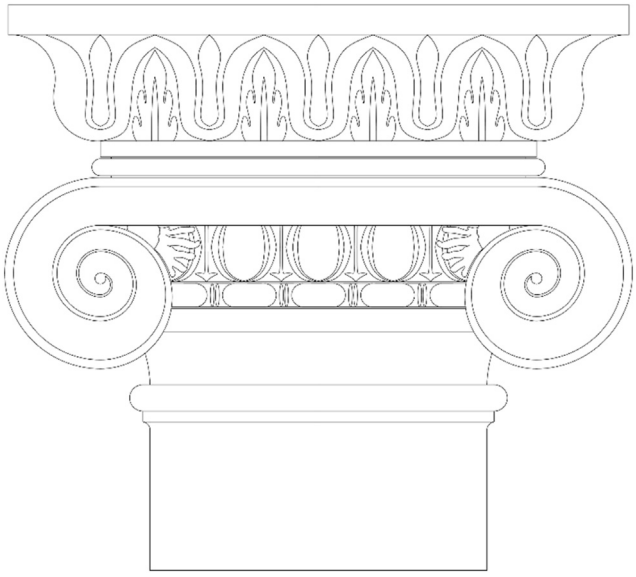


Fig. 16 Representation of one of the ionic capitals that surmounts the small columns of the matronea of the second internal order.

Dizionario Illustrato dei Simboli: Simboli, Emblemi, Attributi, Allegorie, Immagini degli Dei, ecc.).

The oldest known shapes of capitals come from Egypt. By analogy, capitals developed evoking local plant forms, such as lotiform, papyriform, and palmiform capitals. In the Greek world, Corinthians capitals have various turns of acanthus leaves that recall the highest part of the plant.

The Greeks elaborated capitals that followed the three classical orders of their architecture (Doric, Ionic, and Corinthian). The Doric order is the simplest variant of the three orders. This order is characterized by having a quarter-of-a-circle echinus and a collarino under a square abacus. Contemporaneously with the aeolic capital (proto-ionic), the ionic capital takes place that takes up the volutes. Such capital has a low abacus, collarino with astragalus, and echinus with ovolos. This capital developed only in the front view with two volutes. Accordingly, the problem of the capital at an angle is born, which is then solved with the arrangement of diagonals over the volutes at an angle, which gave rise to the successive ionic solution of four volutes and four balusters. The Corinthian capital is characterized by two or three turns of acanthus leaves on an inverted frustoconical bell (calado) and angles occupied by diagonal volutes, which is frequent in hellenistic and roman architecture. The Corinthian capital later merges in Rome with the ionic capital, thereby giving birth to the composite capital (Pevsner, N. et al., 2019. *Dizionario di Architettura*).

Of all these types of capitals, those that are present inside and outside the Baptistery of Florence are the ionic, Corinthian, composite, and heterodox capitals with several variants. The ionic capital is found in the columns of the second internal order (matroneum). This capital is the most evolved with four non-angular volutes, four balusters, an echinus with ovolos, an astragalus with fusarole, a high collarino, and a subtle abacus; the whole is surmounted by a latastrum that evidently evokes the arboreal foliage (Fig. 16).

The Corinthian capital is found in the first internal order, in the columns and the angular pilasters (Photo 7), and externally, in the pilasters of the secondary sides of the first order, in the octagonal columns of the secondary sides of the second order, and in the pilasters that decorate the lateral windows of the main and secondary sides, with the exception of the lateral windows of the two main sides (east and west sides) and of the right capitals of the two lateral windows of one secondary side (southwest side). For example, a typical capital of the cantonal pilasters is composed of an abacus, a tablet with apophyge and a cymatial listel, decorated with a flower, a basket (campana) with three turns of acanthus leaves, two cauliculae with small leaves, and two central helixes (Fig. 17) (Photo 8).



Photo 7 Corinthian capitals: angular pilasters and columns of the first internal order.

The composite capital is found internally, in the columns of the eastern (Photo 9) and northern faces of the first order, and externally, in the columns of the main sides (north, south, and east) of the first order and in the octagonal columns of the three main sides of the second

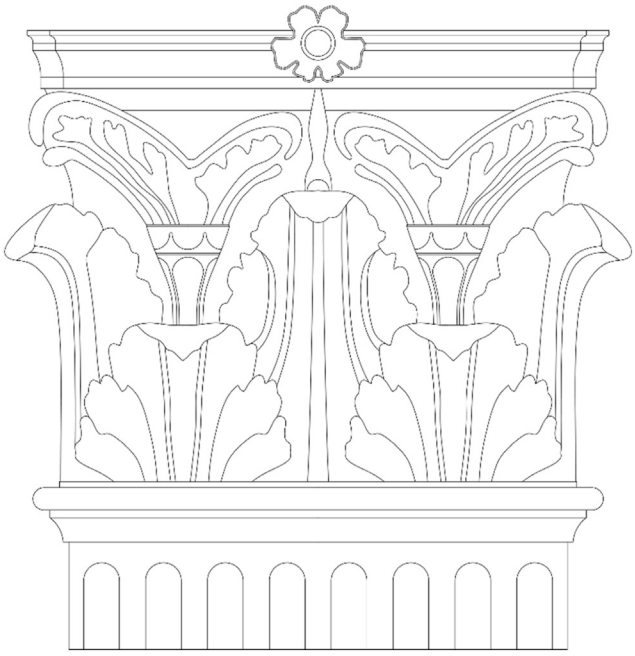


Fig. 17 Representation of one of the Corinthian capitals that surmounts the two cantonal pilasters of the angle formed by the east-northwest sides of the first internal order.



Photo 8 Corinthian capital of one of the angular pilasters of the first internal order.

order. The capitals of the small column-pilasters that decorate the central windows of the main and secondary sides are composite, with the exception of the west side (scarsella). The two right capitals of the pilasters that decorate the two lateral windows on the southwest side are also composite. A typical composite capital is constituted by a falconed abacus with components similar to those of Corinthian capitals, four angular volutes, an echinus, an astragalus similar to those ionic, and a campana with two turns of acanthus leaves. The composite capital of the column that frames the right door of the first internal order lacks an astragalus but has a high head with lithotomy



Photo 9 Capital composed of the column that frames the left side of the east gate of the first internal order.

decoration and corollas, also lithoformed, around the two tubed cauliculae (Fig. 18).

The heterodox capital is found internally, in the pilasters that divide the choirs of the matronea (women's galleries), and externally, in the pilasters that divide the sides of the third order (attic). A typical heterodox capital, such as the one that overcomes the attic pilasters, is made up of a simple abacus (listel and tablet) decorated with a flower or with an infructescence, a campana of variable height with two turns of acanthus leaves, various half-piped or tubed cauliculae, and angular volutes in the vertical as a result of the envelopment in themselves of the second leaf tongue (Fig. 19).

3.5. Entablatures: architraves, friezes, and cornices and types of entablatures

The word architrave comes from the Greek—Latin term that means *trave maggiore*, that is, the element of the entablature that rests on the capitals of the columns, the pilasters, or the estipites.

The symbology of the architrave is linked to that of the square or rectangle because it represents the horizontal trave beam of a trilith above an opening and rests on posts in the form of an elongated square (rectangle). This symbology "would symbolically represent the perfection of the relationship established between earth and heaven and would describe the desire of the members of society to participate in such perfection" (Chevalier, Jean e Gheerbrant, alain. 1996. *Dictionnaire des symbols*). Architraves often cross as protection over the church entrances, just as the bow is a threat for those who dare to enter without being in the mood for it.

We can also talk about architraved entablature, especially when the constituent parts of the structural entablature (architrave, frieze, and cornice) are lined with

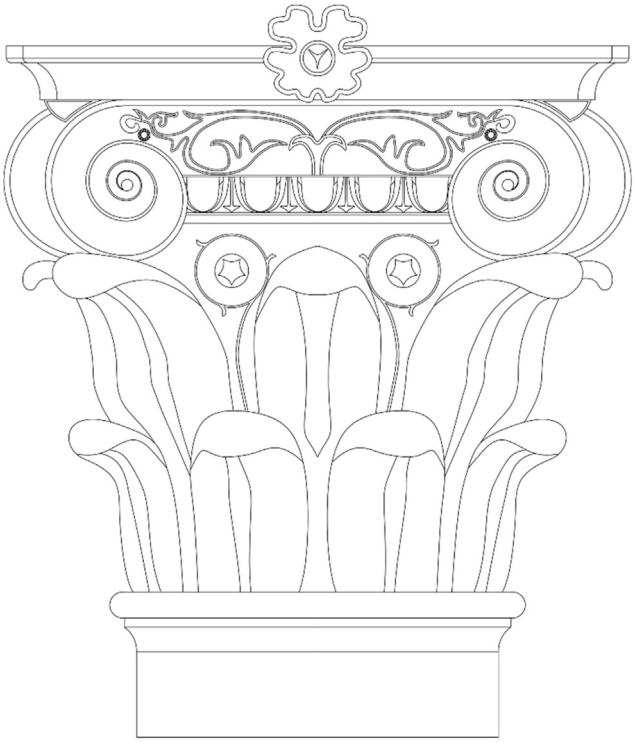


Fig. 18 Representation of the composite capital that surmounts the right column of the north gate of the first internal order.

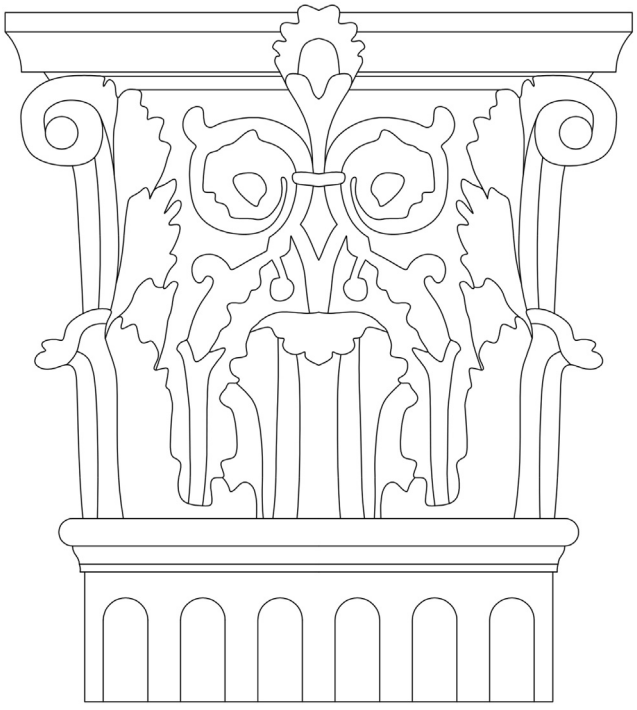


Fig. 19 Representation of one of the heterodox capitals that surmounts the lesenes that divide the sides of the third external order known as attic.

profiled fasciae, such as those of the Baptistery of San Giovanni. Two types of entablature can be recognized within the baptistery: one that separates the first of the second order (midwives) and another that divides the second order (midwives) from the third (false drum).

The entablature of the first internal order is constituted, starting from below, by 1) an architrave with four fasciae, three plain fasciae, and a cyma reversa, separated by three different types of astragali; 2) a mosaicked frieze with cherubs and seraphim; and 3) a cornice formed by a cyma reversa, an echinus with ovolos, a grooved-profile cavetto, and a cyma recta. The constituent elements of the cornice are separated from each other by different fasciae or listels (Photo 10) and (Fig. 20).

The entablature of the second internal order is constituted, starting from the bottom, by 1) an architrave formed by three plain fasciae and a cyma reversa, separated by three similar astragali; 2) a mosaicked frieze similar to that of the first order; 3) a cornice formed, from the bottom up, by an echinus, a cavetto, a fascia with minor dentils, an echinus with ovolos, a second fascia with minor dentils, a second cavetto, and a listel (Photo 11) and (Fig. 21).

"The subdivision of the entablature in the three members of the architrave, frieze and cornice, fixed either by vitruvian theory, or by countless examples of buildings of antiquity, has been substantially taken into account with scrupulous fidelity by the architects who conceived the marble lining of both internal and external claddings of the 'San Giovanni'". If the characteristic profiles of the Corinthian style are punctually kept in the survey of the architrave and frieze moldings, then the big cornice significantly moves away from the typical molding, especially in the substantial elimination of the gutter (Paolucci, Antonio. curated by. Mirabilia Italiae 2: Il Battistero di San Giovanni a Firenze. Salvi, Federica. 1994. *Gli elementi architettonici del Battistero: capitelli, basi e trabeazioni*).

External and internal architraves are similar to a few exceptions. All architraves are formed from top to bottom by: 1) a plain listel; 2) a cyma reversa—in almost all cases—without decoration; 3) a third astragalus; 4) a third plain fascia; 5) a second astragalus; 6) a second plain fascia; 7) a first astragalus—always deprived of decoration; and 8) a first plain fascia. The majority of cymae reversas



Photo 10 Architrave, frieze, and cornice of the entablature of the first internal order.

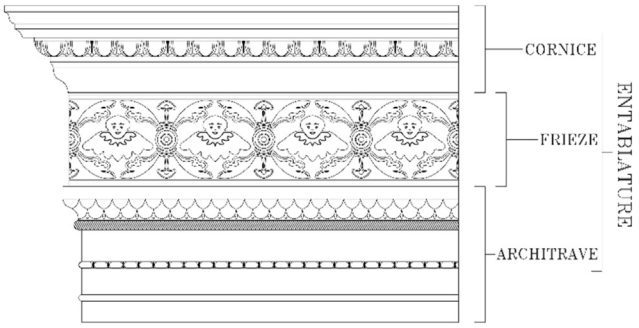


Fig. 20 Representation of the architrave, frieze, and cornice of the entablature of the first internal order.



Photo 11 Architrave frieze and cornice of the entablature of the second internal order.

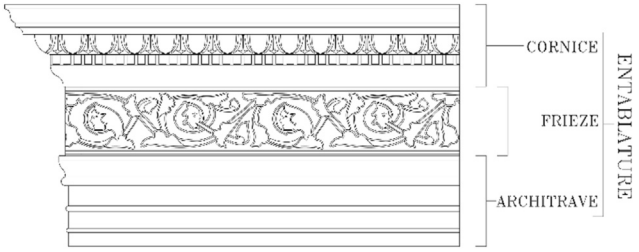


Fig. 21 Representation of the architrave, frieze, and cornice of the entablature of the second internal order.

and astragali have no decoration but in a single exceptional case. The number of plain fasciae is three, also with only one exceptional case (Fig. 22).

The architraves of the first and second internal orders, the first and second external orders, and the third external order (attic) are constituted by the three fasciae (first, second, and third) with the exception of those which, together with the rest of the entablature and the capitals, constitute the impost base of the arcades of the second external order; these architraves have only two fasciae (first and second) (Fig. 23).

The three astragali that crown each of the three fasciae of the architrave lack decoration. Only two astragali of the architrave of the first internal order have decoration: one with tangled rope and the other with fusarole (Fig. 24).

The cymae reversas are almost devoid of decoration, except for that of the first internal order architrave, which has decoration made of scales (Fig. 24).

The architrave is part of the entablature that decorates the windows of the second external order. This part varies depending on the side to which the windows belong. Three types of architraves can be identified. The first type is constituted from top to bottom by 1) a low-height smooth listel known as cymatial, 2) a grooved cavetto, 3) a first fascia a little higher than the listel and the cavetto together, and 4) a second fascia a little less high than the previous one. This first type of architrave is present in the central and lateral windows of the main side (south side) and in the lateral windows of the three secondary sides (northeast, southeast, and southwest sides), with the exception of the right window of the southwest side; the profile has the same characteristics of the second architrave model (Fig. 25).

The second type of architrave is composed from top to bottom by 1) a first low-height smooth listel known as a cymastial, 2) a slightly taller cyma reversa, 3) a first low-height astragalus, 4) a first plain fascia, 5) a second astragalus of equal height as that of the previous one, and 6) a second plain fascia with a shorter in height than that of the previous one. This type of architrave is present in the central and lateral windows of the main side (north side), in the lateral windows of the secondary side (northwest side), and in the right window of the southwest side (Fig. 26).

The third type of architrave is verified only in the central window of the east side and is formed from top to bottom by: 1) a first upper listel known as cymatial, 2) a cyma reversa decorated with double flakes and darts, 3) an astragalus with a tilted rope instead of a second lower listel, 4) a first plain fascia, 5) an astragalus with fusarole; 6) a second plain fascia with a lesser height than that of the previous one, 7) a fascia with smaller teeth and even height than that of the astragalus, and 8) a third plain fascia of lesser height than those of the two previous ones (Fig. 27).

The west side (scarcella) does not have any type of architrave in its windows, given that these windows are not those of the pedestal type. The lateral windows on the east side which, despite of being those of the kneepad type,



Fig. 22 Representation of the architrave of the first external order.



Fig. 23 Representation of the architrave that constitutes the impost of the arcades of the second external order.

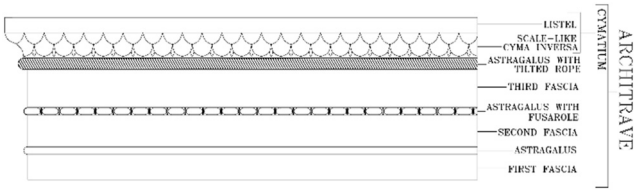


Fig. 24 Representation of the architrave of the first internal order.



Fig. 25 Representation of the first type of architrave of the windows of the second external order.



Fig. 26 Representation of the second type of architrave of the windows of the second external order.

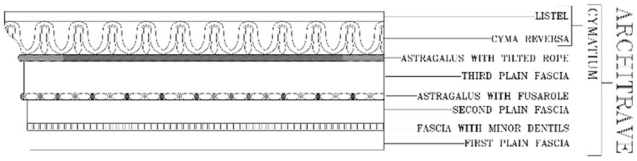


Fig. 27 Representation of the third type of architrave of the windows of the second external order.

have an ornamental semicircular arch instead of the typical entablature.

4. Conclusion

The research results were clearly evidenced through the architectural survey and *relievo* of the constituent elements of the classical architectural orders present in the Baptistery of Florence and the interpretation of their symbolic load based on architectural symbology bibliography research.

The column is structurally the essential element of the Baptistery of Florence architecture and construction. Such column symbolizes the support and axis of the construction that unites its different levels (the first three orders; internal and external). Let us not confuse the "orders" or "levels" of the baptistery that are actually five: the first order is where the three entrance doors are located; the second order or matronea is where the choirs were located; the third order or false drum is the octagonally shaped structural element where the dome rests; the fourth order is the octagonal dome; and the fifth order is known as the "octagonal lantern" that crowns the dome, with the three architectural orders of Greco-Roman classicism: Doric, Ionic, Corinthian, Tuscan, and Composite. The columns guarantee the stability of the religious monument and support the octagonal dome; demolishing them means putting the solidity of the construction at risk. For this reason, the columns are interpreted as an indissoluble whole and part inherent to the building and inconceivable in isolation.

The column with the base, shaft, and capital has a meaning that goes beyond the physical concept of support and, by analogy, denotes the tree of life. The base symbolizes the roots, shaft, trunk, capital, and foliage. By pragmatic extension, the column gives life to the building it supports. The shape of the column is originally inspired by the tree. The columns of the baptistery of Florence are no exception, even though not everyone may interpret it this way. The origin of the column shape dates back to the construction in ancient times when houses were built with tree trunks, which bases were embedded in the earth similar to the tree roots and the ground. The trunk, the vertical element, transmitted the load of the building to ground, as do the concrete columns do, towards the foundations, nowadays. The foliage, the branches of the trunks, was part of the roof structure and served to support the roof made of dry plant fibers as a means of protection from the weather. According to experts, the Roman origin of the Baptistery of Florence was built under the ruins of a Roman structure and initially considered a temple dedicated to Mars. This temple was nothing more than an octagonal building with exterior walls without a roof but with a memorial column to the center, which is a symbol of the tree of life. In the excavations carried out at the beginning of the previous century, a foundation of a Roman plinth that would have served as support for this column was found under the current pavement of the baptistery. However, this theory was questioned in the 700s, maintaining the controversy to this day and the uncertainty of the foundation of the monument itself (IV—V or VII century) (Giusti, Annamaria. 2013. *Il Battistero di San Giovanni a Firenze*).

The elements that form the columns and entablatures lead us to the individual symbolic analysis of the parts. Nevertheless, the relationship of the columns and other vertical support elements, such as small columns and pilasters, also speak after themselves in terms of symbology by either their number, location, or grouping. For example, the eight is at the base of the conception of the baptistery itself with eight sides. Eight is the number of the cosmic equilibrium and cardinal directions. In Christian symbology, eight is the number of eternity, the one that follows the six days of Creation plus Saturday. Eight is present in all five orders or levels of the baptistery. Eight is the number of the grooves of the cantonal pilasters of the first internal order. The two is instead a symbol of opposition and balance. In Christianity, dualism is not only between father and son but also between heaven and earth. Two is the number of columns on each side of the baptistery. Cantonal pilasters are shown in groups of two. Two is also the number of entablatures inside the baptistery. Three is the universal and fundamental number of perfection and stability; it is also the trinity division in Christianity, that is, God, the Son, and

the Holy Spirit. Each side of the baptistery has three intercolumn spaces. The baptistery has three main orders inside (ground floor, matronea, and false drum) and outside (ground floor, matronea, and attic). Three orders of classical architecture are tangible in columns (bases, shafts, and capitals) and entablatures (architraves, friezes, and cornices). Three is also the number of entablatures outside the baptistery. Three is the structural divisions of the three external and internal orders.

The topic of the elements of the Baptistery of Florence symbology may be the stimulus for further studies, as described in the previous paragraph, on numerical symbology, color symbology, symbology of geometric shapes, and symbology of materials, despite the Christian symbolism ascribed that may not be particularly obvious for the observer due to the strong classical order heritage. The survey and relievo have allowed discovering part of the past and the origin of the Baptistery of Florence, thus contributing to its understanding. The experience gained with the application of survey and relievo techniques, plus those of symbolic interpretation, in this study on the classical orders of the Baptistery of Florence, is also valid to point out to similar objectives to study our patrimonial historical assets.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foar.2020.07.002.

References

Accidini Cristina, 1997. In: Illustrata (Ed.), Il Battistero e il Duomo di Firenze. Mondadori Electa, Florence, p. 77.

Busignani, A., Bencini, R., 1988. Le Chiese di Firenze. Il Battistero di San Giovanni. Le Lettere, Florence, p. 28.

Cairo Giovanni, 2001. Dizionario Ragionato dei Simboli. Forni, Bologna, p. 81.

Chevalier Jean e Gheerbrant Alain, 1996a. Dictionnaire des symboles. Editions Robert Laffont S. A. e Editions Jupiter, 1969. Italian translation: Dizionario dei simboli-miti sogni costumi gesti forme figure colori numeri. RCS Rizzoli Libri S., Milano, pp. 295—296.

Chevalier Jean e Gheerbrant Alain, 1996b. Dictionnaire des symboles. Editions Robert Laffont S. A. e Editions Jupiter, 1969. Italian translation: Dizionario dei simboli-miti sogni costumi gesti forme figure colori numeri. RCS Rizzoli Libri S., Milano, p. 298.

Chevalier Jean e Gheerbrant Alain, 1996c. Dictionnaire des symboles. Editions Robert Laffont S. A. e Editions Jupiter, 1969. Traduzione italiana: Dizionario dei simboli-miti sogni costumi gesti forme figure colori numeri. RCS Rizzoli Libri S., Milan. A., 286.

Del Rosso Giovanni, 1820. Ricerche Storiche architettoniche sopra il Singolarissimo Tempio di San Giovanni. Giuseppe Molini e Comp., Florence, pp. 40—41.

Giusti Annamaria, 2013. Il Battistero di San Giovanni a Firenze. Franco Cosimo Panini, Florence, p. 158.

Paolucci Antonio curated by Mirabilia Italiae 2: Il Battistero di San Giovanni a Firenze. Salvi, Federica, 1994. Gli elementi architettonici del Battistero: capitelli, basi e trabeazioni. Modena: Franco Cosimo Panini Editore s, p. 138.

Pevsner N, et al., 2019. *Dizionario di Architettura*. Giulio Einaudi editore s., Turin, p. 118.

Rees Elizabeth, 1994. Simboli Cristiani e Antiche Radici. Cinisello Balsamo, San Paolo, p. 106.

Ronchetti Giuseppe, 1979a. Dizionario Illustrato dei Simboli: Simboli, Emblemi, Attributi, Allegorie, Immagini degli Dei, ecc. Hoepli, Milan, p. 811.

Ronchetti Giuseppe, 1979b. Dizionario Illustrato dei Simboli: Simboli, Emblemi, Attributi, Allegorie, Immagini degli Dei, etc. Hoepli, Milan, p. 406.

Ronchetti Giuseppe, 1979c. Dizionario Illustrato dei Simboli: Simboli, Emblemi, Attributi, Allegorie, Immagini degli Dei, etc. Hoepli, Milan, p. 812.

Sinisgalli Rocco, 2003. Il Battistero di Firenze. Cadmo, Florence, p. 37.

Verdon Timothy, 2017. Il grande museo del Duomo di Firenze. Mandragora, Florence, p. 134.

Verdon Timothy curated by, 1992. Alla Riscoperta di Piazza del Duomo in Firenze. Dal Battistero al Duomo. Florence: Centro Di, p. 44.